

Dr Antonis (Antony) Vamvakeros

Research Engineer / Applied Scientist — Machine Learning for Scientific Data

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Profile

Research scientist with over 10 years of experience developing computational and machine learning methods for large-scale scientific imaging and inverse problems. My work spans physics-informed method development and hands-on experimental research, including the design and execution of cutting-edge operando and multimodal experiments at large-scale facilities.

Experience

Nov 2022 - present: **Royal Society Industry Fellow, Dyson School of Design Engineering, Imperial**

- Developed physics-informed and generative models (GANs, flow-matching) for multimodal imaging, enabling data fusion and spatial resolution enhancement
- Created the [Battery Imaging Library \(BIL\)](#), the first open-access multi-modal and multi-length-scale imaging resource for battery research
- Supervised PhD student on [image segmentation](#) tasks using Meta's SAM and Dino foundation models; selected work was presented at NeurIPS 2024 [AI4Mat Workshop](#)

Feb 2021 - present: **Research & Development Lead Scientist, Finden Ltd**

- Led and delivered multiple competitively awarded UKRI R&D projects (£900k+ total), acting as technical lead and co-PI
- Defined the technical scope and led engineering development of [BeamStop](#), Finden's first commercial software for chemical imaging and tomography data
- Proprietary [PQ-Net](#) code licensed non-exclusively to a diffractometer OEM
- Open-source [nDTomo](#) code adopted by DECTRIS Ltd as the core diffraction imaging analysis tool in their [cloud platform](#)

May 2017 - Feb 2021: **Research Scientist, Finden Ltd**

- Designed and deployed algorithms and machine learning methods for experimental data, including [tomographic reconstruction](#), self-supervised data denoising, [artefact reduction](#), image segmentation and [fast spectroscopic data analysis](#)
- Delivered advanced materials characterisation and machine learning solutions for industrial clients and UK government agencies

Jan 2019 - Jan 2021: **Honorary Research Fellow, Dept. of Chemistry, UCL**

- Developed a new image reconstruction algorithm, termed [Direct Least-Squares Reconstruction](#) algorithm, for chemical tomography data

May 2017 - Jan 2019: **Postdoctoral Scientist, European Synchrotron Radiation Facilities**

- Developed Python-based tools for tomographic reconstruction and data analysis
- Deployed computational tools directly in live beamline experiments with facility users

Education

2017: **PhD in Operando Chemical Tomography**, Chemistry, University College London, UK

2013: **MSc in Chemical Process Engineering**, University College London (Distinction), UK

2012: **MEng in Chemical Engineering**, University of Patras, Greece

Selected Open-Source Code and Data

[nDTomo](#), [BIL](#), [Vulture](#), [HR-Dv2](#), [SAMBA](#), [SD2I](#)

Awards

- Royal Society of Chemistry 2023 Analytical Science Horizon Prize: "The XRD-CT Pioneers"
- Clark Prize, Department of Chemistry, UCL (2016)